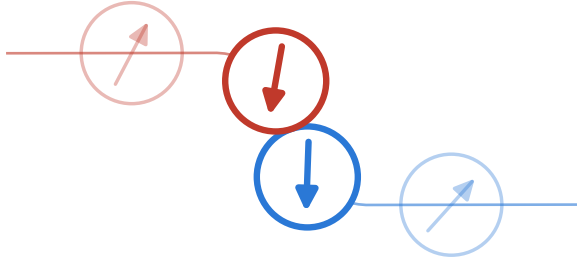
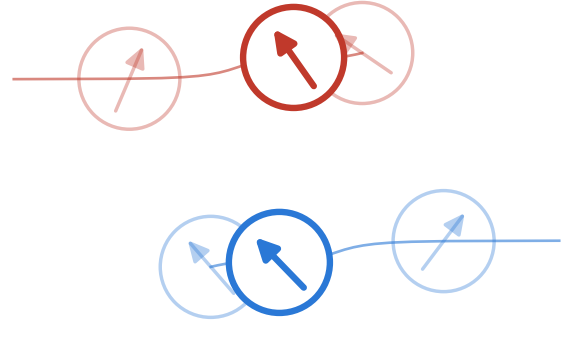
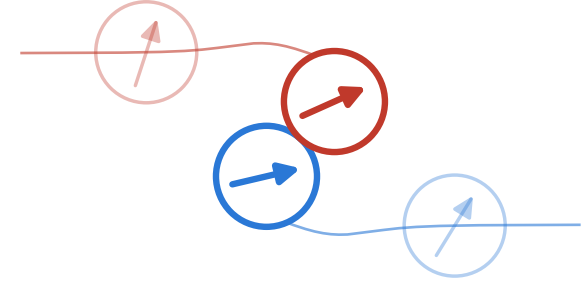
(a) $b = 3.0R$: contact



(b) $b = 3.2R$: escape



(c) $b = 3.4R$: contact